崑山科技大學機械工程系大學部四年制課程科目表

Kun Shan University Department of Mechanical Engineering

4-Year College Curriculum (Day Division)

課程名稱(講授時數—實習時數—學分數)

Course Name (Lecture Hours – Practice Hours –Credits)

112.11.06 系課程委員通過
112.11.15 院課程委員會通過
113.05.13 臨時校課程委員會通過
114.03.04 系課程委員會修正通過
114.04.22 院課程委員會修正通過
114.07.22 校課程委員會修正通過

113 學年度入學適用

Applicable to the freshmen enrolled in Fall, 2024

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)							
校訂共同科目(合計 2 學分) General Courses (2 Credits)							
	生活實踐教育(一) Life Practical Education(I) (0-1-1)	生活實踐教育(二) Life Practical Education(II) (0-1-1)	服務學習與公民教育 Public Service and Civil Education (0-1-0)				
核心通識科目(合計 17 學分) Common Core Courses (17 Credits)							
體育 (一) Physical Education (I) (2-0-2)	體育 (二) Physical Education (I) (2-0-2)			生涯發展與職業倫理 Career Development and Work Ethics (工民商 2-0-2)	外語選讀 Foreign Languages ** (工商 2-0-2)		英文能力檢定* English Proficiency Examination* (1-0-1)
中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2)	中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2)						
英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2)						
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)							
分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)			
8-0-8	8-1-9	2-1-3	2-1-2	4-0-4(工商創)	2-0-2(工商創)		(1-0-1)

註：Note:

*「外語能力檢定」為本校畢業門檻，依「崑山科技大學學生外語基本能力檢定實施辦法」辦理。

'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。

'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

院訂基礎必修科目（合計 15 學分）General Compulsory Courses of College (Total：15 Credits)
（機械系、電機系、電子系、機器人系）

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
微積分(一) Calculus I (2-0-2) 工程技術概論 Introduction to Engineering Technology (2-0-2)	微積分(二) CalculusII (2-0-2) 程式設計 Programming (1-2-3)	工程數學(一) Engineering MathematicsI (3-0-3) 科技英文 Technical English (2-0-2)					學術倫理* Academic ethics (0-1-1)
4-0-4	3-2-5	5-0-5	0-0-0	0-0-0	0-0-0	0-0-0	0-1-1

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

院訂專業模組必修/必選科目(合計 21 學分) Professional Modules Compulsory/Required Courses of College (Total：21 Credits)

機電整合專業模組課程（機械系、電機系、電子系、機器人系）

普通物理與實驗(一) General Physics and Experiment (I) (2-1-3)(必修) 可程式控制器實習 PLC practice (1-2-3) (選修)	機器人工程師實務 Robot Engineer practice (1-2-3) (選修) 電路學 Electric Circuits (3-0-3) (選修)	電子學與實習 Electronics and Practice （1-2-3）(選修) 充電樁系統概論 Introduction to Charging Pile System （3-0-3）(選修)	車用晶片模組應用系統概論 Introduction to Automotive Chip Module Application Systems （3-0-3）(選修)				
2-4-6	4-2-6	4-2-6	3-0-3	0-0-0	0-0-0	0-0-0	0-0-0

智慧車輛組 Intelligent Vehicles

專業必修科目(合計 37 學分) Professional Compulsory Courses (Total: 37 Credits)

圖學 Graphics (1-2-3) 車輛工程 Vehicle Engineering (3-0-3)	普通物理與實驗(二) General Physics and Experiment (II) 靜力學 Statics (3-0-3) 車輛原理及實習(一) Vehicle Principles and PracticeI (1-2-3)	材料力學 Mechanics of Materials (3-0-3) 車輛原理及實習(二) Vehicle Principles and PracticeII (1-2-3) 電腦輔助繪圖 Computer Aided Drawing (1-2-3)	工程數學(二) Engineering MathematicsII (3-0-3) 工程材料學 Engineering Materials (3-0-3) #電動車概論 Introduction to electric vehicles (3-0-3) 熱力學 Thermodynamics (3-0-3)	實務專題(一) HandsOnI (0-1-1)	實務專題(二) HandsOnII (0-1-1)	暑期校外實習 Summer off-campus internship (0-*-2)	機械專業證照 Mechanical professional certificate (0-1-0)
4-2-6	4-2-6	5-4-9	12-0-12	0-1-1	0-1-1	0-*-2	0-1-0

專業選修科目 Professional Elective Courses

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
普通化學 General Chemistry (3-0-3)	車輛先進科技概論 Introduction to Advanced Vehicle Technology (2-0-2)	車輛專業英文 Vehicle professional English (2-0-2) 動力學 Dynamics (3-0-3) 機械加工實習(一) Machining PracticeI (0-2-2) 內燃機 Internal combustion engine (3-0-3) &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2) &機聯網證照輔導 IOT certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2)	車聯網概論 Introduction to Internet of Vehicles (2-0-2) 車輛綜合檢修 Comprehensive vehicle maintenance (1-2-3) 創意思考與方法 Creative thinking and methods (2-0-2) 機械加工實習(二) Machining PracticeII (0-2-2) 電子學暨實習(二) Electronics and practiceII (1-2-3) 機構學 Mechanisms (3-0-3) 自動變速箱原理與設計 Automatic transmission principle and design (3-0-3) 基礎量測及實習 Fundmental Measurement and Practice (1-2-3)	車輛動力學 Vehicle dynamics (3-0-3) 流體力學 Fluid mechanics (3-0-3) 車輛機構設計 Vehicle mechanism design (2-0-2) 電動車馬達系統與實習 Electric vehicle motor system and practice (1-2-3) 自動控制 Automatic control (3-0-3) 機械元件設計 Mechanical component Design (3-0-3) 車輛感測技術 Vehicle sensing Technology （3-0-3） *車用晶片模組規劃與設計 Automotive chip module planning and design （3-0-3）(必選) *車用晶片模組製造生產實務 Automotive chip module manufacturing and production practices （3-0-3）(必選) *車用晶片模組測試檢驗實務 Automotive Chip Module Testing and Inspection Practice （2-2-4）(必選)	油電混合車技術 Hybrid vehicle technology （3-0-3） 高壓電池系統與實習 High Voltage Battery Systems and practice (1-2-3) 車用燃料學 Automotive Fuel Science (3-0-3) 車輛振動學 Vehicle Vibration （3-0-3） 車輛省能與防污技術 Vehicle energy saving and Anti-fouling technology (3-0-3) 燃料電池應用技術 Fuel cell application technology (3-0-3) 氣液壓技術及實習 Pneumatic and hydraulic technology and practice (3-0-3) #智慧車輛控制系統 Smart vehicle control system (3-0-3) 保養廠營運與管理 Maintenance shop operation and management （2-0-2）	電動車機電整合及實習 Electric vehicle electromechanical integration and practice (1-2-3) 車輛 CANBUS 實務技術 Vehicle CANBUS practical technology (3-0-3) 引擎動力計及實習 Engine Dynamics and practice (1-2-3) 電腦工程應用及實習 Computer Engineering Applications and practices (1-2-3) 軌道車概論 Introduction to rail cars (2-0-2) 模具開發實務 Mold development practice (3-0-3) 校外實習 Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internshipI (0-*-9)	電動車動力系統檢修實務 Electric vehicle power system maintenance practice (1-2-3) 引擎設計 Engine design (3-0-3) 汽車行銷學 Automotive Marketing (2-0-2) 振動與噪音防治 Vibration and noise prevention (3-0-3) 工業管理 Industrial Management (2-0-2) 學年校外實習(二) Academic year off-campus internshipII (0-*-9) 校外職場實習 Off-campus workplace internship (0-*-7)

***分組必選課程** Mandatory elective course

智慧製造組 Smart Manufacturing

專業必修科目(合計 40 學分) Professional Compulsory Courses (Total: 40 Credits)

圖學 Graphics (1-2-3) 機械加工實習 Machining PracticeI (一) (0-3-3)	普通物理與實驗(二) General Physics and Experiment (II) (2-1-3) 靜力學 Statics (3-0-3) 機械製圖 Mechanical Drawing (1-2-3)	材料力學 Mechanics of Materials (3-0-3) 動力學 Dynamics (3-0-3) #機構學 Mechanisms (3-0-3)	工程數學(二) Engineering MathematicsII (3-0-3) 熱力學 Thermodynamics (3-0-3) 機械設計 Mechanical Design (3-0-3)	實務專題(一) HandsOn I (0-1-1) 熱工實驗 Thermal engineering Lab (1-2-3)	實務專題(二) HandsOnII (0-1-1)	暑期校外實習 Summer off-campus internship (0-*-2)	機械專業證照 Mechanical professional certificate (0-1-0)

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
1-5-6	6-3-9	9-0-9	9-0-9	1-3-4	0-1-1	0-*-2	0-1-0
專業選修科目 Professional Elective Courses							
普通化學 General Chemistry (3-0-3)	工具機概論 Introduction to machine tools (3-0-3) 機械加工實習 Machining PracticeII (二) (0-3-3)	流固力實驗 Fluid and Solid Mechanics Lab (0-3-3) 氣液壓實習 Pneumatic and hydraulic practice （1-2-3） 應用性機構組立與實習 Assemble of applied Mechanism (1-2-3) 材料學與實驗 Materials Science & Experiments (1-2-3) 2D 電腦繪圖實習 2D computer drawing practice (1-2-3) CNC 加工乙級實習(一) CNC machining Grade B PracticeI （1-2-3）	流體力學 Fluid Mechanics (3-0-3) 熱處理 Heat treatment (3-0-3) 精密軸承設計 Precision bearing design (3-0-3) 實物測繪實習 Physical mapping Practice （1-2-3） 3D 電腦繪圖實習 3D computer drawing practice (1-2-3) CNC 加工乙級實習(二) CNC machining Grade B PracticeII （1-1-2） #電腦輔助繪圖 Computer Aided Drawing (1-2-3) 基礎量測及實習 Fundmental Measurement and Practice (1-2-3) 電子學暨實習(二) Electronics and practiceII (1-2-3)	磨潤工程 Tribology engineering (3-0-3) 節能技術導論 Introduction to Energy Saving Technology (3-0-3) 工業產品設計 Industrial product design (3-0-3) 創意思考與方法 Creative thinking and methods (2-0-2) 3D 造型設計實習 3D modeling design practice (1-2-3) 智慧製造生產管理 Smart manufacturing production management (3-0-3) #進階數控加工與實習 Advanced CNC machining and practice （1-2-3） #工業 4.0 特色技術 Industry 4.0 featured technologies (3-0-3) &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2)	非傳統加工 Non-traditional machining (3-0-3) 自動控制 Automatic control (3-0-3) 機械工程專題討論 Mechanical Engineering Seminar (1-0-1) 基礎科學數位教學與實習 Basic science digital teaching and practice （1-2-3） 基礎力學數位教學與實習 Basic mechanics digital teaching and practice （1-2-3） 自動化機械系統實習 Automated Mechanical Systems Practice （1-2-3） #智慧製造技術 Smart manufacturing technology (3-0-3) &機聯網證照輔導 IOT certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2)	複合材料 Composite materials (3-0-3) 校外實習 Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internshipI (0-*-9) 模具開發實務 Mold development practice (3-0-3) 工業管理 Industrial Management (2-0-2)	學年校外實習(二) Academic year off-campus internshipII (0-*-9)
精密機械組 Precision machinery							
專業必修科目(合計 40 學分) Professional Compulsory Courses (Total: 40 Credits)							
圖學 Engineering Graphics (1-2-3) 機械加工實習(一) Machining Practice I (0-3-3)	普通物理與實驗(二) General Physics and Experiment (II) (2-1-3) 靜力學 Statics (3-0-3) 機械製圖 Mechanical Drawing (1-2-3)	材料力學 Mechanics of Materials (3-0-3) 動力學 Dynamics (3-0-3) #機構學 Mechanisms (3-0-3)	工程數學(二) Engineering MathematicsII (3-0-3) 熱力學 Thermodynamics (3-0-3) 機械設計 Mechanical Design (3-0-3)	PBL 專題製作(一) PBL HandsOn I (0-1-1) 熱工實驗 Thermal engineering Lab (1-2-3)	PBL 專題製作(二) PBL HandsOn II (0-1-1)	暑期校外實習 Summer off-campus internship (0-*-2)	機械專業證照 Mechanical professional certificate (0-1-0)
1-5-6	**6-3-9**	**9-0-9**	**9-0-9**	**1-3-4**	**0-1-1**	**0-*-2**	**0-1-0**
專業選修科目 Professional Elective Courses							

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
普通化學 General Chemistry (3-0-3) 綠能科技與實作 Green energy technology and practice (2-1-3)	綠色科技概論 Introduction to Green Technology (2-0-2) 綠色應用概論 Introduction to Green Applications (2-0-2) 工程材料學 Engineering Materials (3-0-3) 機械加工實習(二) Machining Practice II (0-3-3)	機械製造 Mechanical Manufacture (3-0-3) 機電儀控整合及實習 Mechanical and electrical instrument control integration and practice (1-2-3) CNC 加工乙級實習(一) CNC machining Grade B PracticeII （1-2-3）	公差原理設計 Tolerance principle design (3-0-3) 太陽能工程概論 Introduction to Solar Engineering (3-0-3) 高等工程數學 Advanced Engineering Mathematics (3-0-3) 節能技術 Energy saving technology (3-0-3) 熱傳學 Thermal transfer (3-0-3) 太陽光電技術 Solar photovoltaic technology (2-0-2) 就業技能輔導 (0-2-0) Employment skills counseling 機械元件設計 (3-0-3) Mechanical component design (3-0-3) #電腦輔助繪圖 Computer Aided Drawing (1-2-3) 基礎量測及實習 Fundmental Measurement and Practice (1-2-3)	系統化創新設計方法 Systematic innovative design method (3-0-3) 自動控制 Automatic control (3-0-3) 物聯網與大數據於智慧製造應用 IOT and Big Data in Smart Manufacturing Applications (3-0-3) 電腦輔助設計 Computer aided design (1-2-3) 流體力學 Fluid mechanics (3-0-3) &CNC 工具機證照輔導 CNC machine tool certificate Counseling (0-*-2) &機聯網證照輔導 IOT certificate Counseling (0-*-2) &感知器工程師證照輔導 Sensor engineer certificate Counseling (0-*-2) 電子學暨實習(二) Electronics and practiceII (1-2-3)	感測器原理與應用 Sensor Principles and Applications (1-2-3) 有限元素法及實習 Finite element method and practice (1-2-3) 氣液壓技術及實習 Pneumatic and hydraulic technology and practice (3-0-3) 流固力實驗 Fluid Mechanics Lab (1-2-3) 創意思考與方法 Creative thinking and methods (2-0-2) 機械工程專題討論 Mechanical Engineering Seminar (1-0-1) CPS 製造技術 CPS manufacturing technology (3-0-3) 鍋爐與廢熱回收 Boilers and waste heat recovery (3-0-3) 空調工程 Air conditioning project (2-0-2)	精密機械設計實務 Precision Machinery Design Practice (3-0-3) 熱傳學概論 Introduction to Heat Transfer (2-0-2) 數值分析及實習 Numerical analysis and practice (1-2-3) 感測器設計 Sensor design (2-0-2) 設備減震與防噪 Equipment shock absorption and noise prevention (3-0-3) 模具開發實務 Mold development practice (3-0-3) 校外實習 Off-campus internship (0-*-2) 學期校外實習 Semester off-campus internship (0-*-9) 學年校外實習(一) Academic year off-campus internshipI (0-*-9)	機器手臂進階 Robot arm advancement (1-2-3) 創意性機構設計 Creative mechanism design (3-0-3) 伺服系統設計 Servo system design (3-0-3) 機器動力學 Machine dynamics (2-0-2) 精密機械傳動元件技術 Precision mechanical transmission component technology (3-0-3) 高等工程分析 Advanced Engineering Analysis (3-0-3) 學年校外實習(二) Academic year off-campus internshipII (0-*-9) 校外職場實習 Off-campus workplace internship (0-*-7) 燃燒學 Combustion science (3-0-3) 奈米工程概論 Introduction to Nanoengineering (2-0-2) 工程論文導讀 Introduction to engineering papers (2-0-2)

最低畢業學分：130 學分 Minimum for total credits: 130 Cs

電子與量測數位科技微學程(各組皆合計 8 學分)

Digital Technology Micro-course (Total: 8Credits)

智慧車輛組 Intelligent Vehicles		
程式設計 Programming (1-2-3)	電子學與實習 Electronics and Practice (1-2-3)	通識(E 類) Liberal Education(E) (2-0-2)
智慧製造組 Smart Manufacturing		
程式設計 Programming (1-2-3)	電子學與實習 Electronics and Practice (1-2-3)	通識(E 類) Liberal Education(E) (2-0-2)
精密機械組 Precision machinery		
程式設計 Programming (1-2-3)	電子學與實習 Electronics and Practice (1-2-3)	通識(E 類) Liberal Education(E) (2-0-2)

電動車跨院系學分學程(合計 15 學分)智慧車輛組適用

Cross-Disciplinary Electric Vehicle Academic Program (Total: 15Credits)

電動車概論 (3 學分) Introduction to electric vehicles (3Credits)	智慧車輛控制系統 (3 學分) Smart vehicle control system (3Credits)	機電儀控整合及實習(3 學分) 機電儀控整合及實習 Mechanical and electrical instrument control integration and practice (3Credits)	綠色能源 (3 學分) Green Energy (3Credits) (外系-學程科目) Courses from Other Departments	綠色產業 (3 學分) Green Industry (3Credits) (外系-學程科目) Courses from Other Departments

智能化製造跨院系學分學程(合計 15 學分)智慧製造組適用

Cross-Departmental Program in Intelligent Manufacturing (Total: 15Credits)

電腦輔助繪圖 (3 學分) Computer Aided Drawing (3Credits)	精密量測及實習 (3 學分) Dimensional Metrology and Practice (3Credits)	機構學 (3 學分) Mechanisms (3Credits)	綠色能源 (3 學分) (外系-學程科目) Green Energy (3Credits) (Courses from Other Departments)	綠色產業 (3 學分) (外系-學程科目) Green Industry (3Credits) (Courses from Other Departments)

機電整合跨院系學分學程(合計 15 學分)精密機械組適用

Mechatronics Interdisciplinary Program(Total: 15Credits)

電腦輔助繪圖 (3 學分) Computer Aided Drawing (3Credits)	精密量測及實習 (3 學分) Dimensional Metrology and Practice (3Credits)	機構學 (3 學分) Mechanisms (3Credits)	綠色能源 (3 學分) (外系-學程科目) Green Energy (3Credits) (Courses from Other Departments)	綠色產業 (3 學分) (外系-學程科目) Green Industry (3Credits) (Courses from Other Departments)

學生須修畢至少任一跨院系學程(上述機械系自訂、或工學院或他院之跨院系學分學程)始得畢業。

Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.

註：1. 暑期校外實習、校外實習課程皆於於暑期實施，前 160 小時為必修，可抵暑期校外實習 2 學分，第 161-320 小時為選修，可抵校外實習 2 學分；實習期滿成績及格者方給予承認課程並抵免學分。「學期校外實習」課程須於同一機構全職實習連續至少 18 週(不可分次實施) 。「學年校外實習(一)、(二)」課程須於同一機構全職實習連續至少 36 週。以上校外實習課程另訂有作業要點，修習前須依辦法提出申請。另外，學期校外實習或學年校外實習(二)可抵暑期校外實習必修 2 學分，與校外職場實習選修 7 學分。

Both the summer off-campus internship and the off-campus internship courses are implemented during the summer vacation. The first 160 hours are compulsory and can be credited for 2 credits of the summer off-campus internship (compulsory). Hours 161-320 are elective and can be credited for 2 credits of the off-campus internship (elective). Course recognition and credit exemption will only be granted to those who successfully complete the internship and pass the assessment. The 'Semester Off-Campus Internship' course requires full-time internship at the same institution for at least 18

consecutive weeks (cannot be done in separate periods). The 'Academic Year Off-Campus Internship (I) and (II)' courses require full-time internship at the same institution for at least 36 consecutive weeks. Operational guidelines for the aforementioned off-campus internship courses are separately stipulated, and applications must be submitted in accordance with the regulations before taking the courses. Furthermore, the semester off-campus internship or the academic year off-campus internship (II) can be credited for the 2 compulsory credits of the summer off-campus internship, and 7 elective credits of the off-campus workplace internship.

2. 有@標記之課程為教育部推動大專院校非資通訊系所之「電子與量測數位科技微學程」課程，需選修通識中心所開(通識類別 E)數位科學領域課程 2 學分及本系所開 6 學分，合計修滿 8 學分可獲頒此微學程證書。本系因應此微學程所開設之課程如下：
   a. 智慧車輛組、機密機械組：程式設計(3 學分)與電子學及實習(3 學分)。
   b. 智慧製造組：程式設計(3 學分)與電子學及實習(3 學分)。

   Courses marked with '@' are part of the 'Electronic and Measurement Digital Technology Microprogram' promoted by the Ministry of Education for departments outside of the information and communication technology (ICT) fields. Students are required to take 2 credits of digital science domain courses (General Education Category E) offered by the Center for General Education and 6 credits offered by our department, totaling 8 credits, to be awarded this microprogram certificate. The courses offered by our department for this microprogram are as follows:
   a. Smart Vehicles Group, Precision Machinery Group: Programming (3 credits) and Electronics and Lab (3 credits).
   b. Intelligent Manufacturing Group: Programming (3 credits) and Electronics and Lab (3 credits)."
3. 有&標記的是彈性密集式證照輔導課程，上課時段將集中於考照前 1-2 個月實施密集式上課，選課前請詳閱系辦選課公告。
   Courses marked with '&' are flexible intensive license preparatory courses. Class times will be concentrated in intensive sessions conducted 1-2 months before the certification exam. Please refer to the department office's course registration announcement before registering for the course.